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Week3 - Literature Review

• Background/Motivation:
– Describe the context in which the research was conducted.
This research was conducted in the context of wage projections being valuable in the job market but being difficult to do precisely. The author emphasizes that machine learning is a method that may make wage predictions more accurate.

– What problem or gap in the existing literature does the paper aim to address?
The existing literature builds machine learning models and compares models to determine which has the best accuracy. This paper goes beyond that and implements or deploys their models into an interactive tool where users can enter their information and actually see what level their salary or wages should be at given their variables. This takes the paper from just a theoretical study to a real life solution.

– Discuss the significance of the research question and why it is important.
Salary prediction is significant because it allows people to make informed decisions or appropriate negotiations when it comes to accepting job offers. Information on salaries is not easily accessible, so creating an accurate way to calculate salaries based on user features would help tremendously in providing fair wages. Companies can also use this information as well. By creating an easily accessible standard to compare to, companies can assess the market price of job roles and determine if hiring additional help is affordable or not.

• Methods Used:
– Summarize the methodologies employed by the authors.
This paper uses web scraping techniques to collect 6,704 observations from the web, and trained supervised machine learning models on this dataset that included multiple variables for each observation such as age, gender, education, job role, experience, and salary as the target. The supervised machine learning algorithms used were linear regression, polynomial regression, support vector regression, decision trees, random forests, gradient boosting, k-nearest neighbors, and neural multilayer perceptron.

– Explain how these methods are suited to address the research question.
What made these methods suitable was the exploratory data analysis that used heatmaps and line plots that proved strong correlations between salary and some of the variables. This made regression-based machine learning techniques appropriate for this dataset.

– Discuss any innovative approaches or techniques that are particularly noteworthy.
This paper used standard machine learning techniques, but it was innovative in that it applied the findings from these techniques to an interactive web application.

• Significance of the Work:
– Highlight the key findings and contributions of the paper.
The random forest model had the highest accuracy, with 91.9%, followed by KNN then Decision trees. Linear Regression had the lowest accuracy of 65.2%, followed by Polynomial regression and Neural Networks.

According to
the EDA, the most important variables were age and years of experience, having the most correlation with salary; while gender and job title have the weakest correlation with salary.

– Explain why the results are important within the broader context of the field.
The results are important because they show which machine learning models are the most effective at predicting salaries, which provides a solid foundation for stakeholders in the labor market to make decisions based on a standard salary level. With the interactive web application, job seekers can now engage in accurate salary negotiations and employers can now keep up with salary expectations for different employees based on labor cost in the market.

– Discuss the implications of the findings for future research or practice.
These findings can create an additional tool for career guidance. Users can plan ahead by providing the tool their information and what education level they would want to attain and have the model determine what their salary would be. This, along with creating a standard salary range for users to fairly negotiate with employers are possible implications of these findings. Future research could include adding more features like location and industry to the models so users can have more accurate information based on their situation.

• Connection to Other Work:
– Relate the paper to other relevant studies.
This paper builds on earlier salary prediction research that applied supervised machine learning algorithms. All but one of the other papers mentioned found that random forest was the most accurate model in their analysis. This validates the strength of random forest for salary prediction compared to other supervised models.

– How does this paper build on or differ from previous work?
This paper builds on previous work by using pretty much the same core models used in previous papers. It differs in its data collection method because it used web scraping techniques to pull data from linkedin for its dataset. It also attached an interactive tool for users to use the model to predict their salary by putting in their information.

– Identify any references to seminal works or influential papers cited by the authors.
The paper references several influential papers:
Bangdiwala (2018) on linear regression as a statistical foundation for prediction models.
Belgiu & Drăguţ (Random Forest in remote sensing) for justification of Random Forest's strength in handling complex, high-dimensional data.
Shukla & Dave (2019) and Mane & Joshi (2020) on comparative machine learning approaches for employee salary
prediction.

• Relevance to Capstone Project:
– Discuss how the content of the paper might be relevant to your own capstone project.
This research is using methodologies I intend to test out and is using them on similar variables (occupational variables) as I am also considering doing.

– Identify any specific methods, theories, or findings that you might incorporate into your project.
As with the last paper I reviewed, the research found that Random Forest was the most accurate model. This reinforces my belief that this would be a great place to start in my project.

– Highlight any potential areas where your capstone could expand upon or diverge from the paper's findings.
My project will involve more features. This paper was limited in features because of the needs of the web application but my project will be able to handle more features such as location, industry, etc.